BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions) (17)

Maximum Marks: 17

Note: Select the correct answer for each from the given options. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 85

Q1. Select the correct answer for each from the given options:

Qi. Kinetic energy per mole of an ideal gas is:

(A) 1/2 kT
(B) 2/3 kT
(C) 3/2 RT
(D) 2/3 RT

Qii. An instrument which can measure and compare potentials without drawing any current from the circuit is called:

(A) Ammeter
(B) Voltmeter
(C) Potentiometer
(D) AV Ometer

Qiii. In laser, the lifetime of an electron in a metastable state is:

(A) 10^{-3} sec
(B) 10^{-8} sec
(C) 10^{3} sec
(D) 10^{9} sec

Qiv. On equipotential surface, work done in moving charged particle is:

(A) Positive
(B) Negative
(C) Zero
(D) Infinite

Qv. Balmer series of hydrogen atom spectrum lies in this region:

(A) Infrared
(B) Ultraviolet
(C) Visible
(D) Radio-wave

Qvi. If the temperature of a cold body is decreased, the efficiency of Carnot engine will:

(A) remain constant
(B) increase
(C) decrease
(D) be zero

Qvii. The device is used to make the path of ionizing particles visible:

(A) Geiger Muller Counter
(B) Cyclotron
(C) Wilson Cloud Chamber
(D) Solid State Detector

Qviii. The unit of potential gradient is:

(A) volt/meter
(B) Coulomb/meter
(C) Volt / Ampere
(D) Volt/Ampere

Qix. If 1A current flows through 2m long wire, the charge flows through it in 2 hours will be:

(A) 3600C
(B) 7200C
(C) 1C
(D) 600 C

Qx. To obtain p-type semiconductor silicon must be doped with the atom whose valency is:

(A) 2
(B) 3
(C) 4
(D) 5

Qxi. Reverse process of photoelectric effect is:

(A) Pair production
(B) Compton effect
(C) Annihilation of matter
(D) X-Ray production

Qxii. Compton effect proves the:

(A) Photon Theory of Light
(B) Dual Nature of Light
(C) Nature of Light
(D) Uncertain Nature of Light

Qxiii. The energy radiated from black body is directly proportional to the fourth power of absolute temperature' This law is:

(A) Wein's Law
(B) Rayleigh Jeans Law
(C) Stefan's Law
(D) Plank's Law

Qxiv. A charge of 0.1C is accelerated through a potential difference of 1000 V, its kinetic energy will be:

(A) 200J
(B) 10J
(C) 100J
(D) 100 eV

Qxv. It is the most suitable material for making core of electromagnet:

(A) Air
(B) Steel
(C) Copper and Nickel Alloy
(D) Soft Iron

Qxvi. coulomb/volt is equal to:

(A) Newton
(B) Joule
(C) Farad
(D) Ohm

Qxvii. Rest mass of photon is:

(A) Infinite
(B) Zero
(C) 9.11×10^{-31}kg
(D) 1.67×10^{-27}kg

Section 'B' (Short-Answer Questions) (36)

Maximum Marks: 36

> Note: Answer any NINE part questions from this section. All questions carry equal marks. Draw diagrams where necessary.

Q2.

(i) Find the shortest and longest wavelength of emitted photons in hydrogen spectra in Paschen series.

(ii) What is Wheatstone Bridge? Prove that for a balanced Wheatstone Bridge: P/Q = R/X.

(iii) X-rays of wavelength 2.64×10^{-10} m are used in Compton scattering process. Find the fractional change in wavelength for a scattering angle of 120°.

(iv) Derive an expression for the force experienced by a current carrying conductor in a uniform magnetic field.

(v) The half-life of ${}_{104}Po^{210}$ is 140 days. By what percent does its activity will decrease per week?

(vi) Calculate the temperature at which the root mean square speed of nitrogen molecule is $3300 ms^{-1}$.

(vii) What is meant by Rectification? How does a semiconductor diode work as a half wave rectifier?

(viii) Give the construction and working of Geiger Counter.

(ix) A particle of charge 3.2×10^{-19} C and mass 2.48×10^{-27} kg is held motionless between two horizontal parallel plates separated by 5cm. Find the potential difference between the plates.

(x) Define Electric Flux. Derive the expression for the electric flux due to a point charge.

(xi) What is thermostat? Describe the working of bimetallic thermostat.

(xii) Find the speed at which mass of particle will be doubled.

(xiii) Three resistances each of 6Ω can be connected in four different ways. What is equivalent resistance of each combination?

(xiv) An iron core solenoid with 400 turns has a cross-section area of 4.0cm². A current of 2.0 ampere passing through it produces B = 0.2 weber/m². What emf is produced in it if the current is turned off in 0.1 sec? What is its self-inductance?

Section "C" (Detailed Answer Questions) (32)

Maximum Marks: 32

> Note: Attempt any TWO questions from this section. All questions carry equal marks. Draw diagrams where necessary.

Q3.

(a) What Photoelectric Effect? Explain its important results. Derive Einstein's photoelectric equation.

(b) Explain capacitance of a capacitor and give its SI unit. Derive the expressions for the capacitance of a parallel plate capacitor when space between the plates contains: i) Free space ii) A dielectric

Q4.

(a) Define Radioactivity. Explain the law of radioactive decay. Write the equations of the change in the parent nuclei in α, β and γ decay.

(b) State the first law of thermodynamics and explain following on its basis: i) Isobaric Process ii) Isochoric Process iii) Isothermal Process iv) Adiabatic Process

Q5.

(a) Derive the expression for the following when $r_n = n_2(4\pi\varepsilon_0 h^2/me^2)$: i) The energy of electron in the nth orbit of hydrogen atom. ii) The wavelength of photons emitted in the hydrogen spectrum.

(b) What is Transformer? On what principle it works? With the help of diagram give its construction and working. Derive expressions for its efficiency and turns ratio.